

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 12: Morphological Operations

WORKSHEET SOLUTIONS

1. 1. Define erosion.

Erosion is a fundamental morphological operation that shrinks or thins binary objects in an image by removing foreground pixels at object boundaries based on the shape of a structuring element.

2. 2. Define dilation.

Dilation is a morphological operation that expands or thickens binary objects in an image by adding foreground pixels to object boundaries based on the shape of a structuring element.

3. 3. Which operation removes small noise?

Erosion (specifically, an **Opening operation**—which consists of erosion followed by dilation) removes small foreground noise pixels while preserving the main shape structure.

4. 4. What is a structuring element?

A structuring element is a small binary matrix/kernel (of predefined shape and size, such as a square, cross, or disk) used to probe an input image during morphological operations to define local neighborhood relationships.